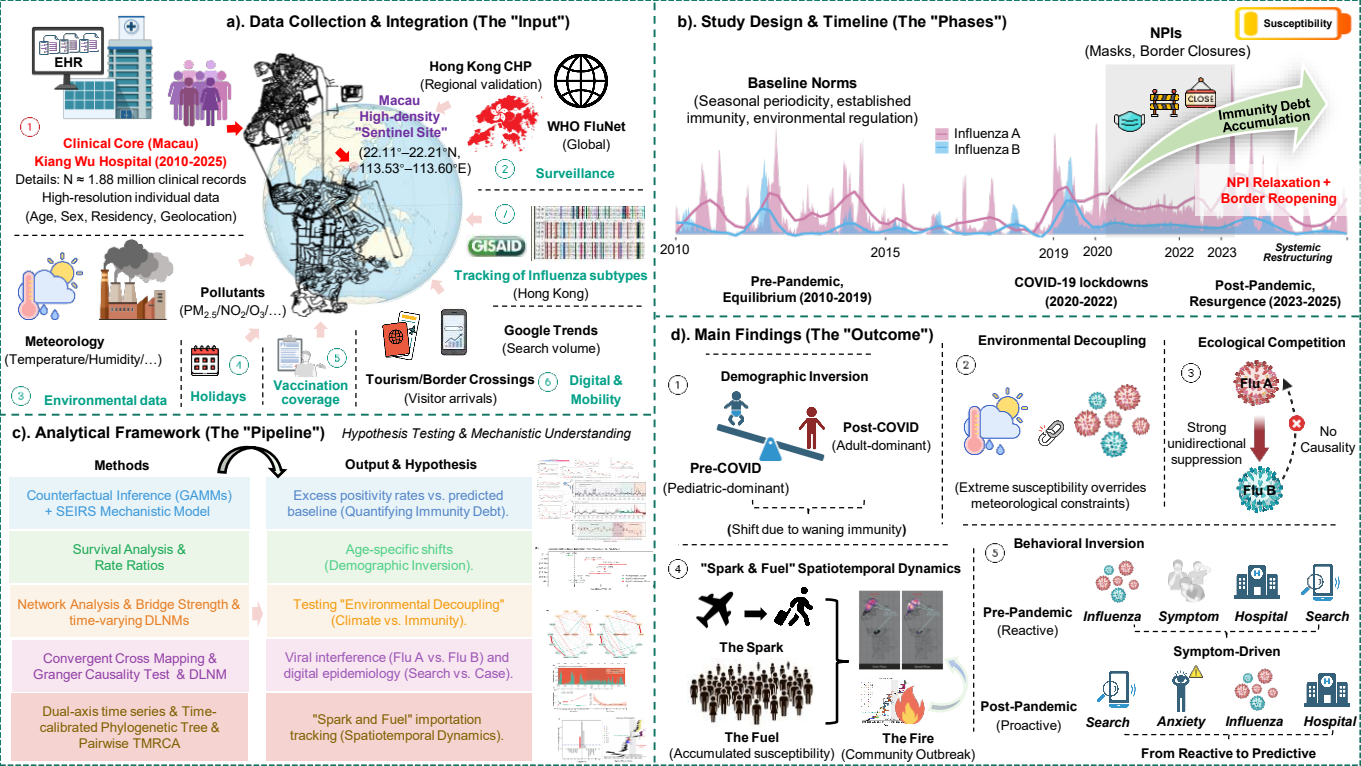

a). Data Collection & Integration (The "Input")
EHR
1
Clinical Core (Macau)
Kiang Wu Hospital (2010-2025)
Details: N ≈ 1.88 million clinical records
High-resolution individual data
(Age, Sex, Residency, Geolocation)
Meteorology
(Temperature/Humidity/…)
3
Environmental data
Pollutants
($PM_{2.5}$/NO_2/O_3/…)
4
Holidays
5
Vaccination coverage
Macau
High-density "Sentinel Site"
(22.11°–22.21°N, 113.53°–113.60°E)
Hong Kong CHP
(Regional validation)
WHO FluNet
(Global)
2
Surveillance
7
GISAID
Tracking of Influenza subtypes
(Hong Kong)
Google Trends
(Search volume)
Tourism/Border Crossings
(Visitor arrivals)
6
Digital & Mobility
b). Study Design & Timeline (The "Phases")
Susceptibility
NPIs
(Masks, Border Closures)
Baseline Norms
(Seasonal periodicity, established immunity, environmental regulation)
Influenza A
Influenza B
Immunity Debt Accumulation
NPI Relaxation + Border Reopening
2010
2015
2019
2020
2022
2023
Systemic Restructuring
Pre-Pandemic, Equilibrium (2010-2019)
COVID-19 lockdowns (2020-2022)
Post-Pandemic, Resurgence (2023-2025)
c). Analytical Framework (The "Pipeline")
Hypothesis Testing & Mechanistic Understanding
Methods
Output & Hypothesis
Counterfactual Inference (GAMMs) + SEIRS Mechanistic Model
Excess positivity rates vs. predicted baseline (Quantifying Immunity Debt).
Survival Analysis & Rate Ratios
Age-specific shifts (Demographic Inversion).
Network Analysis & Bridge Strength & time-varying DLNMs
Testing "Environmental Decoupling" (Climate vs. Immunity).
Convergent Cross Mapping & Granger Causality Test & DLNM
Viral interference (Flu A vs. Flu B) and digital epidemiology (Search vs. Case).
Dual-axis time series & Time-calibrated Phylogenetic Tree & Pairwise TMRCA
"Spark and Fuel" importation tracking (Spatiotemporal Dynamics).
d). Main Findings (The "Outcome")
1
Demographic Inversion
Pre-COVID
(Pediatric-dominant)
Post-COVID
(Adult-dominant)
(Shift due to waning immunity)
2
Environmental Decoupling
(Extreme susceptibility overrides meteorological constraints)
3
Ecological Competition
Flu A
Flu B
Strong unidirectional suppression
No Causality
4
"Spark & Fuel" Spatiotemporal Dynamics
The Spark
The Fuel
(Accumulated susceptibility)
The Fire
(Community Outbreak)
5
Behavioral Inversion
Pre-Pandemic
(Reactive)
Influenza
Symptom
Hospital
Search
Symptom-Driven
Post-Pandemic
(Proactive)
Search
Anxiety
Influenza
Hospital
From Reactive to Predictive